

Multiple Choice

1. **Answer:** A

Options: A) the tRNA transcript from the original DNA; B) a single strand of the original DNA segment after a substitution mutation; C) the final processed mRNA made from the original DNA; D) the primary RNA transcript (after splicing) from the original DNA; E) a single strand of the original DNA segment after a duplication mutation

Note: Correct option: A - the tRNA transcript from the original DNA

2. **Answer:** A

Options: A) Plants, animals, and fungi all have both chloroplasts and mitochondria.; B) Plants and fungi have chloroplasts but no mitochondria; animals have only mitochondria.; C) Animals have both chloroplasts and mitochondria; plants and fungi have only chloroplasts.; D) Plants and animals have chloroplasts but no mitochondria; fungi have only mitochondria.; E) Plants and animals have mitochondria but no chloroplasts; fungi have only chloroplasts.

Note: Correct option: A - Plants, animals, and fungi all have both chloroplasts and mitochondria.

3. **Answer:** A

Options: A) hydrogen; B) methane; C) nitrogen; D) sulfur dioxide; E) neon

Note: Correct option: A - hydrogen

4. **Answer:** A

Options: A) Enzymatic reactions with oxygen; B) Lactose crystallization; C) Loss of refrigeration power over time; D) mesophiles; E) Natural separation of milk components without bacterial influence

Note: Correct option: A - Enzymatic reactions with oxygen

5. **Answer:** D

Options: A) Cue, Desire, Fulfillment; B) Arousal, Behavior, Consequence; C) Stimulus, Habituation, Extinction; D) Appetitive behavior, Consummatory act, Quiescence; E) Perception, Decision, Movement

Note: Correct option: D - Appetitive behavior, Consummatory act, Quiescence

True / False

6. **Answer:** True

Options: A) True; B) False

Note: Bedtime NPH insulin added to maximal therapy with sulfonylurea and metformin is an effective, simple, well-tolerated approach for patients with uncontrolled type 2 diabetes.

7. **Answer:** True

Options: A) True; B) False

Note: Human pWAT has chemotactic properties through the secretion of different chemokines, and we propose that pWAT might contribute to the progression of obesity-associated atherosclerosis.

8. **Answer:** False

Options: A) True; B) False

Note: There is little evidence that the 2008 public antibiotic campaigns were effective. The use and visibility of future campaign materials needs auditing. A carefully planned approach that targets the public in GP waiting rooms and through clinicians in consultations may be a more effective way of improving prudent antibiotic use.

9. **Answer:** True

Options: A) True; B) False

Note: The real-time PCR approach revealed promising results in pollen identification and quantification, even when analyzing pollen mixes. Future perspectives could concern the development of multiplex real-time PCR for the simultaneous detection of different taxa in the same reaction tube and the application of high-throughput molecular methods.

10. **Answer:** False

Options: A) True; B) False

Note: The new storage can affords more stable temperature levels when compared to the formerly used can. Since temperature is stable during conservation, continuous monitoring in everyday practice does not seem warranted.

Long Form Response

11. **Answer:** Robin, Lizard. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** stabilizing selection. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key